

OLOVYANNAYA, Russian Federation

WMO: 309610

Lat: 50.950N

Lon: 115.583E

Elev: 640

StdP: 93.87

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.2	-32.6	-37.6	0.1	-34.9	-35.1	0.1	-32.2	11.8	-18.0	9.4	-19.5	1.0	150	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.6	18.6	29.3	18.0	27.2	17.4	20.9	27.3	19.8	26.1	18.8	24.5	4.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.7	24.0	17.7	13.7	22.5	16.7	12.9	21.5	64.1	27.2	59.8	26.5	56.3	24.4	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.4	8.0	-38.8	35.5	3.2	3.0	-41.1	37.6	-43.0	39.4	-44.9	41.1	-47.2	43.3		
			-38.8	23.5	3.2	3.1	-41.1	25.8	-43.0	27.6	-44.8	29.4	-47.2	31.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-23.5	-18.6	-8.4	3.1	11.2	18.1	20.8	18.0	11.1	1.4	-11.5	-21.2	(6)
(7)		DBStd	16.34	5.14	5.48	6.65	5.07	4.71	4.13	3.13	3.55	4.54	4.94	6.50	5.68	(7)
(8)		HDD10.0	4585	1038	802	570	214	41	1	0	1	39	268	646	966	(8)
(9)		HDD18.3	6817	1297	1036	828	457	226	52	10	49	220	524	896	1224	(9)
(10)		CDD10.0	989	0	0	0	7	79	244	335	248	73	3	0	0	(10)
(11)		CDD18.3	180	0	0	0	0	6	46	87	38	3	0	0	0	(11)
(12)		CDH23.3	1749	0	0	0	3	88	511	779	322	46	0	0	0	(12)
(13)		CDH26.7	585	0	0	0	1	26	186	271	92	9	0	0	0	(13)
(14)	Wind	WSAvg	3.2	2.3	2.5	3.3	4.3	4.6	3.4	3.0	3.3	3.3	2.9	2.2	(14)	
(15)	Precipitation	PrecAvg	335	4	3	5	13	24	63	84	73	33	11	9	6	(15)
(16)		PrecMax	575	13	12	18	41	55	185	257	154	100	26	27	16	(16)
(17)		PrecMin	185	0	0	0	0	2	5	27	21	4	0	1	1	(17)
(18)		PrecStd	99	3	3	5	12	15	49	52	41	24	8	6	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.1	-1.1	11.6	21.9	29.9	35.9	34.9	32.3	27.8	18.3	6.1	-4.5	(19)
(20)			MCWB	-9.6	-4.2	3.9	9.4	14.5	19.8	20.4	19.5	15.9	9.4	1.7	-6.3	(20)
(21)		2%	DB	-11.5	-4.8	7.3	17.6	25.6	31.1	32.0	29.2	24.1	14.3	2.3	-8.7	(21)
(22)			MCWB	-12.4	-6.8	1.2	7.9	13.2	17.2	19.9	18.6	14.9	7.0	-1.1	-9.8	(22)
(23)		5%	DB	-14.1	-7.6	4.4	14.3	22.6	28.4	29.6	26.8	21.2	11.4	0.0	-11.3	(23)
(24)			MCWB	-14.9	-9.1	-0.5	5.6	12.0	17.1	18.6	17.8	13.4	5.2	-2.4	-12.2	(24)
(25)		10%	DB	-16.3	-10.0	1.4	11.6	20.0	26.1	27.5	24.6	18.9	8.9	-2.5	-13.6	(25)
(26)			MCWB	-16.9	-11.2	-2.4	4.2	10.6	16.3	18.3	17.0	12.4	3.8	-4.6	-14.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.5	-4.0	4.2	10.2	16.1	22.3	23.8	21.8	17.7	10.0	1.9	-6.3	(27)
(28)			MCDWB	-8.2	-1.3	10.7	20.2	24.1	30.4	29.9	27.3	24.8	17.5	5.6	-4.6	(28)
(29)		2%	WB	-12.4	-6.8	1.5	8.2	14.2	19.8	21.4	19.9	15.7	7.6	-0.8	-9.8	(29)
(30)			MCDWB	-11.5	-4.9	6.8	16.9	23.7	27.4	27.5	25.9	21.6	13.4	2.0	-8.8	(30)
(31)		5%	WB	-14.8	-9.0	-0.4	6.2	12.7	18.3	20.2	18.8	14.4	5.7	-2.5	-12.2	(31)
(32)			MCDWB	-14.1	-7.5	4.0	13.6	21.3	25.3	26.6	24.2	19.8	11.0	0.0	-11.5	(32)
(33)		10%	WB	-16.9	-11.1	-2.3	4.5	11.1	17.2	19.4	17.9	13.1	4.1	-4.5	-14.2	(33)
(34)			MCDWB	-16.3	-10.0	1.2	11.2	19.2	24.0	25.3	23.0	17.9	8.5	-2.6	-13.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	10.7	11.1	11.1	12.1	11.2	10.4	10.2	11.2	10.1	9.0	8.2	(35)
(36)			MCDBR	9.7	12.1	13.0	15.2	16.1	14.8	13.9	13.9	14.8	13.6	10.8	9.5	(36)
(37)			MCWBR	8.9	10.5	8.5	8.7	7.7	6.2	5.3	6.0	7.6	8.2	8.1	8.6	(37)
(38)		5% WB	MCDWB	9.9	12.2	12.7	14.5	14.6	12.2	11.3	10.6	12.8	12.8	10.8	9.5	(38)
(39)			MCWBR	9.0	10.5	8.6	8.7	7.8	6.4	5.7	5.7	7.3	8.2	8.2	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.278	0.317	0.411	0.486	0.462	0.420	0.387	0.342	0.307	0.277	0.257	(40)	
(41)		taud	2.228	2.182	2.169	2.040	1.883	2.072	2.265	2.336	2.398	2.400	2.288	2.189	(41)	
(42)		Ebn at Noon	781	849	876	822	773	799	831	843	847	814	736	697	(42)	
(43)		Edh at Noon	68	95	118	152	187	156	127	112	94	77	65	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.57	2.74	4.29	5.30	6.01	6.32	5.81	5.10	4.05	2.73	1.59	1.19	(44)	
(45)		RadStd	0.07	0.13	0.20	0.35	0.35	0.50	0.40	0.34	0.29	0.15	0.07	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page